ECO 362: Financial Investments
Arbitrage

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An example from baseball

- In 2016, Stephen Strasburg signed a 7-year contract extension with the Washington Nationals worth $175 million.
- In baseball contracts, salary is guaranteed regardless of performance or injury.
- Significant risk for the team if a player is injured.

Insurance

- Pro Financial Services is a company that underwrites insurance policies on baseball players.
- Two future states of nature:
  1. Strasburg misses more than 60 days during the 2017 season due to injury.
  2. Strasburg misses fewer than 60 days.
- An insurance policy on Strasburg has a random payoff depending on which of the two states is realized.

$$X = \begin{cases} X(1) = 18.3\text{m} & \text{if state 1} \\ X(2) = 0 & \text{if state 2} \end{cases}$$

- Insurance costs $P = 1.3\text{m}$ before the 2017 season.

A simple model of stocks

- Think about stocks in a similar way.
- Current stock price is $P = 1$.
- Future price is

$$X = \begin{cases} X(1) = 1.20 & \text{if bull market} \\ X(2) = 0.84 & \text{if bear market} \end{cases}$$

- Can add more states to make the payoffs more realistic.

State-contingent assets

- Represent any asset with an uncertain future payoff as a state-contingent security.
- Index future states by $s = 1, \dots, S$.
- The probabilities of these states are $\pi(1), \dots, \pi(S)$.
- Index assets by $i = 1, \dots, I$.
- Future payoff X_i of security i is a random variable that takes the value $X_i(s)$ in state s.
- We denote the price of security i today as

$$P_i = P(X_i)$$

- $P(\cdot)$ is a discounting function that transforms a random future payoff to its value today.

Law of one price

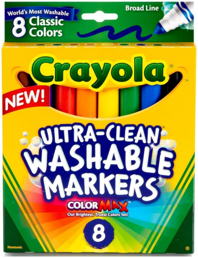
- A portfolio is a linear combination of assets whose random payoff is

$$X = \sum_{i=1}^{I} w_i X_i$$

  where w_i can take any value.
  - $w_i = 1$: Long 1 unit of security i.
  - $w_i = -1$: Short 1 unit of security i.
- The law of one price says that the price of the portfolio is the sum of the prices of the assets in the portfolio.

$$P(X) = P\left(\sum_{i=1}^{I} w_i X_i\right) = \sum_{i=1}^{I} w_i P(X_i)$$

A violation of the law of one price



<u>Amazon Prime</u>
$6.99 per box
$48.77 for a 10 pack

- Potential arbitrage?
  - Set up a store on Amazon, but may be difficult to profit because of shipping costs.
  - There are 200 families at my daughter's school who need a box. I can order 10 packs to cover all families.
- Like Newton's laws, law of one price holds only in ideal conditions without frictions (transaction costs).
- Works better in financial markets than most other markets.

A simple financial market

- Asset 1 (riskless): $P_1 = 0.99$.

$$X_1 = \begin{cases} X_1(1) = 1 & \text{if state 1 (bull)} \\ X_1(2) = 1 & \text{if state 2 (bear)} \end{cases}$$

- Asset 2 (risky): $P_2 = 1$.

$$X_2 = \begin{cases} X_2(1) = 1.20 & \text{if state 1 (bull)} \\ X_2(2) = 0.84 & \text{if state 2 (bear)} \end{cases}$$

Represent as a binominal tree

Asset 1

$$P_1 = 0.99 \quad \begin{cases} X_1(1) = 1 \\ X_1(2) = 1 \end{cases}$$

Asset 2

$$P_2 = 1 \quad \begin{cases} X_2(1) = 1.20 \\ X_2(2) = 0.84 \end{cases}$$

Price any other asset by no arbitrage

- Asset 3 has payoffs

$$X_3 = \begin{cases} X_3(1) = 0.20 & \text{if state 1 (bull)} \\ X_3(2) = 0 & \text{if state 2 (bear)} \end{cases}$$

- What is its price?
  1. Construct a portfolio of assets 1 and 2 that have the same payoffs as asset 3.
  2. Asset 3 must have the same price as the portfolio by no arbitrage.

Portfolio replication

w_1 units of asset 1

$0.99w_1$

$1w_1$

$1w_1$

w_2 units of asset 2

$1w_2$

$1.20w_2$

$0.84w_2$

w_1 units of asset 1 + w_2 units of asset 2 = asset 3

$0.99w_1 + 1w_2$

$1w_1 + 1.20w_2 = 0.20$

$1w_1 + 0.84w_2 = 0$

Portfolio replication

- Construct a portfolio of assets 1 and 2 that have the same payoffs as asset 3.

$$
\begin{aligned}
X_1(1)w_1 + X_2(1)w_2 =& 1w_1 + 1.20w_2 = 0.20 \\
X_1(2)w_1 + X_2(2)w_2 =& 1w_1 + 0.84w_2 = 0
\end{aligned}
$$

- The solution is

$$
\begin{aligned}
w_1 =& -0.47 \\
w_2 =& 0.56
\end{aligned}
$$

- The price of the portfolio is

$$P_1 w_1 + P_2 w_2 = 0.99 \times -0.47 + 1 \times 0.56 = 0.09$$

Arbitrage if law of one price fails

- \$0.09 must be the price of asset 3 if there is no arbitrage.
- If asset 3 were \$0.10, make a \$0.01 profit by
  - Sell asset 3.
  - Buy the portfolio: -0.47 units of asset 1 and 0.56 units of asset 2.
- If asset 3 were \$0.08, make a \$0.01 profit by
  - Buy asset 3.
  - Sell the portfolio: 0.47 units of asset 1 and -0.56 units of asset 2.
- Arbitrage is a riskless profit.

Pure state-contingent assets

State 1-contingent asset | State 2-contingent asset

- State prices are the prices of pure state-contingent assets.

Solving for the state 1 price

- Construct a portfolio of assets 1 and 2 that pays \$1 in state 1.

$$
\begin{aligned}
X_1(1)w_1 + X_2(1)w_2 =& 1w_1 + 1.20w_2 = 1 \\
X_1(2)w_1 + X_2(2)w_2 =& 1w_1 + 0.84w_2 = 0
\end{aligned}
$$

- The solution is

$$
\begin{aligned}
w_1 =& -2.33 \\
w_2 =& 2.78
\end{aligned}
$$

- By no arbitrage,

$$
\begin{aligned}
m(1) =& P_1 w_1 + P_2 w_2 \\
=& 0.99 \times -2.33 + 1 \times 2.78 = 0.4678
\end{aligned}
$$

Solving for the state 2 price

- Construct a portfolio of assets 1 and 2 that pays $1 in state 2.

$$
\begin{aligned}
X_1(1)w_1 + X_2(1)w_2 =& 1w_1 + 1.20w_2 = 0 \\
X_1(2)w_1 + X_2(2)w_2 =& 1w_1 + 0.84w_2 = 1
\end{aligned}
$$

- The solution is

$$
\begin{aligned}
w_1 =& 3.33 \\
w_2 =& -2.78
\end{aligned}
$$

- By no arbitrage,

$$
\begin{aligned}
m(2) =& P_1 w_1 + P_2 w_2 \\
=& 0.99 \times 3.33 + 1 \times -2.78 = 0.5222
\end{aligned}
$$

Intuition for state prices

- State prices are

$$m(1) = 0.4678 \text{ for state 1 (bull)}$$
$$m(2) = 0.5222 \text{ for state 2 (bear)}$$

- State price for the bear market is higher.
- \$1 in a bear market (e.g., high unemployment, high inflation, raging war,...) is more valuable than \$1 in a bull market.
- Thus, assets that deliver higher payoffs in worse states are more valuable.

An alternative solution for the price of asset 3

- Asset 3 has the same payoffs as
  - 0.2 units of state 1-contingent asset.
  - 0 units of state 2-contingent asset.
- By no arbitrage, the price of asset 3 is

$$P_3 = \sum_{s=1}^{2} m(s)X_3(s)$$
$$= 0.4678 \times 0.20 + 0.5222 \times 0 = 0.09$$

Pricing any other asset

- Suppose that asset 4 has payoffs

$$X_4 = \begin{cases} X_4(1) = 0.20 & \text{if state 1 (bull)} \\ X_4(2) = 0.16 & \text{if state 2 (bear)} \end{cases}$$

- By no arbitrage, the price of asset 4 is

$$\begin{aligned} P_4 =& \sum_{s=1}^{2} m(s) X_4(s) \\ =& 0.4678 \times 0.20 + 0.5222 \times 0.16 = 0.18 \end{aligned}$$

Stochastic discount factor

- Define the stochastic discount factor as $M(s) = m(s)/\pi(s)$, which is high for
  - More valuable states: high $m(s)$.
  - Less likely states: low $\pi(s)$.
- The price of any asset i is

$$
\begin{aligned}
P_i &= \sum_{s=1}^{2} m(s) X_i(s) \\
&= \sum_{s=1}^{2} \pi(s) M(s) X_i(s) = \mathbb{E}[M X_i]
\end{aligned}
$$

- There are only 2 states in this example, but the same math generalizes to an arbitrary number of states.

Adding a time dimension

- Our model so far has only two dates, today and future.
- The future could be any future date: 1 year, 2 years,....
- Let's add time subscripts (and drop the i subscript for simplicity).

$$P = \mathbb{E}[M_t X_t]$$

- M_t is the stochastic discount factor for discounting a risky payoff at date t.

Present-value formula for risky assets

- An asset has random payoffs X_1 at date 1, X_2 at date 2,...
- The law of one price implies that the price of the asset is the sum of the prices of the payoffs at different dates.
- Thus, the price of the asset is

$$P = \sum_{t=1}^{\infty} \mathbb{E}[M_t X_t]$$

Riskless asset as a special case

- A riskless asset has payoff of 1 in all future dates and states: $X_1 = 1$, $X_2 = 1$,...
- Applying the present-value formula,

$$P = \sum_{t=1}^{\infty} \mathbb{E}[M_t \times 1]$$

- From the lecture on "Discounting", the present-value formula for a riskless payoff is

$$P = \sum_{t=1}^{\infty} \frac{1}{(1 + Y_b(t))^t} = \sum_{t=1}^{\infty} P_b(t)$$

  where $Y_b(t)$ is the yield and $P_b(t)$ is the price of a t-year zero-coupon bond.
- Thus, the expectation of the stochastic discount factor is the price of a zero-coupon bond.

$$\mathbb{E}[M_t] = P_b(t)$$

A probability refresher

- Random variable X with mean $\mathbb{E}[X]$ and standard deviation $\sigma(X)$.
- Random variable Y with mean $\mathbb{E}[Y]$ and standard deviation $\sigma(Y)$.
- The covariance between X and Y is

$$\begin{aligned}\mathrm{Cov}(X, Y) =& \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\ =& \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]\end{aligned}$$

- The correlation between X and Y is

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma(X)\sigma(Y)} \in [-1, 1]$$

Idiosyncratic vs. systematic risk

1. Idiosyncratic: $\mathrm{Cov}(X_i, M) = 0$.
   - A risk that is diversifiable.
   - No correlation across payoffs.
   - Becomes riskless if you have a large portfolio of payoffs.
2. Systematic: $\mathrm{Cov}(X_i, M) \neq 0$.
   - A risk that is non-diversifiable.
   - Correlation across payoffs.
   - Risk remains even if you have a large portfolio of payoffs.

Idiosyncratic or systematic?

- Is Strasburg's risk of injury idiosyncratic or systematic?
  - Idiosyncratic if injuries are uncorrelated across baseball players.
  - Even if injuries are correlated, they may be diversified with other types of disability insurance.
- What about flood insurance in NJ?
  - Idiosyncratic if floods are local events that can be diversified across states and countries.
  - Floods may also be diversified with other natural disasters.
  - However, global warming may make floods more likely around the world (and possibly affect the global economy).
- Most macro risk is systematic: Growth and inflation in the US, Fed's monetary policy, trade wars, etc.

Present-value formula for any idiosyncratic payoff

- Rewrite the present-value formula as

$$
\begin{aligned}
P &= \sum_{t=1}^{\infty} \mathbb{E}[M_t X_t] \\
&= \sum_{t=1}^{\infty} (\mathbb{E}[M_t]\mathbb{E}[X_t] + \mathrm{Cov}(M_t, X_t)) \\
&= \sum_{t=1}^{\infty} \left(\frac{\mathbb{E}[X_t]}{(1 + Y_b(t))^t} + \mathrm{Cov}(M_t, X_t) \right)
\end{aligned}
$$

- For an idiosyncratic payoff,

$$
P = \sum_{t=1}^{\infty} \frac{\mathbb{E}[X_t]}{(1 + Y_b(t))^t}
$$

- Use the zero-coupon yield curve to discount any idiosyncratic payoff (not just constant ones).

Apply the formula to the baseball example

- The Strasburg insurance is an example of risky but idiosyncratic payoff.
- In December 2016, the 1-year zero-coupon yield is 0.9%.
- Let π be the probability of injury (i.e., missing more than 60 days).
- Apply the present-value formula.

$$1.3 = \frac{\pi 18.3 + (1-\pi)0}{1.009}$$

  which implies that $\pi = 7.17\%$.
- Pro Financial Services's estimate of injury probability must be no more than 7.17%.
- Good business as long as the pool of insured players is sufficiently large (i.e., law of large numbers applies).

An alternative representation of present value

- An asset has random payoffs $X_{i,1}$ at date 1, $X_{i,2}$ at date 2,...
- Instead of holding the asset forever, you can sell it at price $P_{i,1}$ after collecting the payoff $X_{i,1}$ at date 1.
- Your rate of return from date 0 to 1 is

$$1 + R_i = \frac{P_{i,1} + X_{i,1}}{P_{i,0}}$$

- Write the present-value formula as

$$\begin{aligned} P_{i,0} =& \mathbb{E}[M(X_{i,1} + P_{i,1})] \\ 1 =& \mathbb{E}[M(1 + R_i)] \end{aligned}$$

- For a 1-year zero-coupon bond,

$$1 = \mathbb{E}[M(1 + Y_b(1))]$$

Asset pricing formula

- Taking the difference between the previous two equations,

$$
\begin{aligned}
0 =& \mathbb{E}[M(R_i - Y_b(1))] \\
=& \mathbb{E}[M]\mathbb{E}[R_i - Y_b(1)] + \mathrm{Cov}(M, R_i - Y_b(1)) \\
=& \mathbb{E}[M]\mathbb{E}[R_i - Y_b(1)] - \mathrm{Cov}(-M, R_i)
\end{aligned}
$$

- Rewrite the previous equation as

$$
\mathbb{E}[R_i - Y_b(1)] = \frac{\mathrm{Cov}(-M, R_i)}{\mathbb{E}[M]}
$$

  - Higher $\mathrm{Cov}(-M, R_i)$ means higher risk because the asset delivers low return when the stochastic discount factor is high.
  - Higher risk is compensated through higher expected returns $\mathbb{E}[R_i - Y_b(1)]$.
- We will come back to this concept in Chapter 5 (CAPM).

Summary

- Present-value formula: Discount any stream of cash flows by the stochastic discount factor.
- Based on the law of one price, which may not hold exactly in practice.
- Asset pricing formula: Tradeoff between risk and return.

Next steps

- To understand the fundamental determinants of expected returns on risky assets, we need a theory of supply-demand.
- The demand for risky assets comes from a portfolio-choice problem. Investors trade off risk and return.
- How to construct the stochastic discount factor in actual data?